POLIMI GRADUATE SCHOOL OF MANAGEMENT

BOSTON airbnb LISTINGS ANALYSIS

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airbnb

OUTLINE

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4. DESCRIPTIVE ANALYSIS
5. MULTIPLE LINEAR REGRESSION (**MLR**)
6. GENERALIZED LINEAR MODEL (**GLM**)
7. LIMITATION
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INTRODUCTION

Project Goal:
Identify determinants of airbnb listing competitiveness.

Dependent Variable (Y)
Number of Reviews per month (proxy for listing competitiveness).

DATA CARD

To make the data digestible, we grouped our 95 variables into these categories:

PRICING: Price

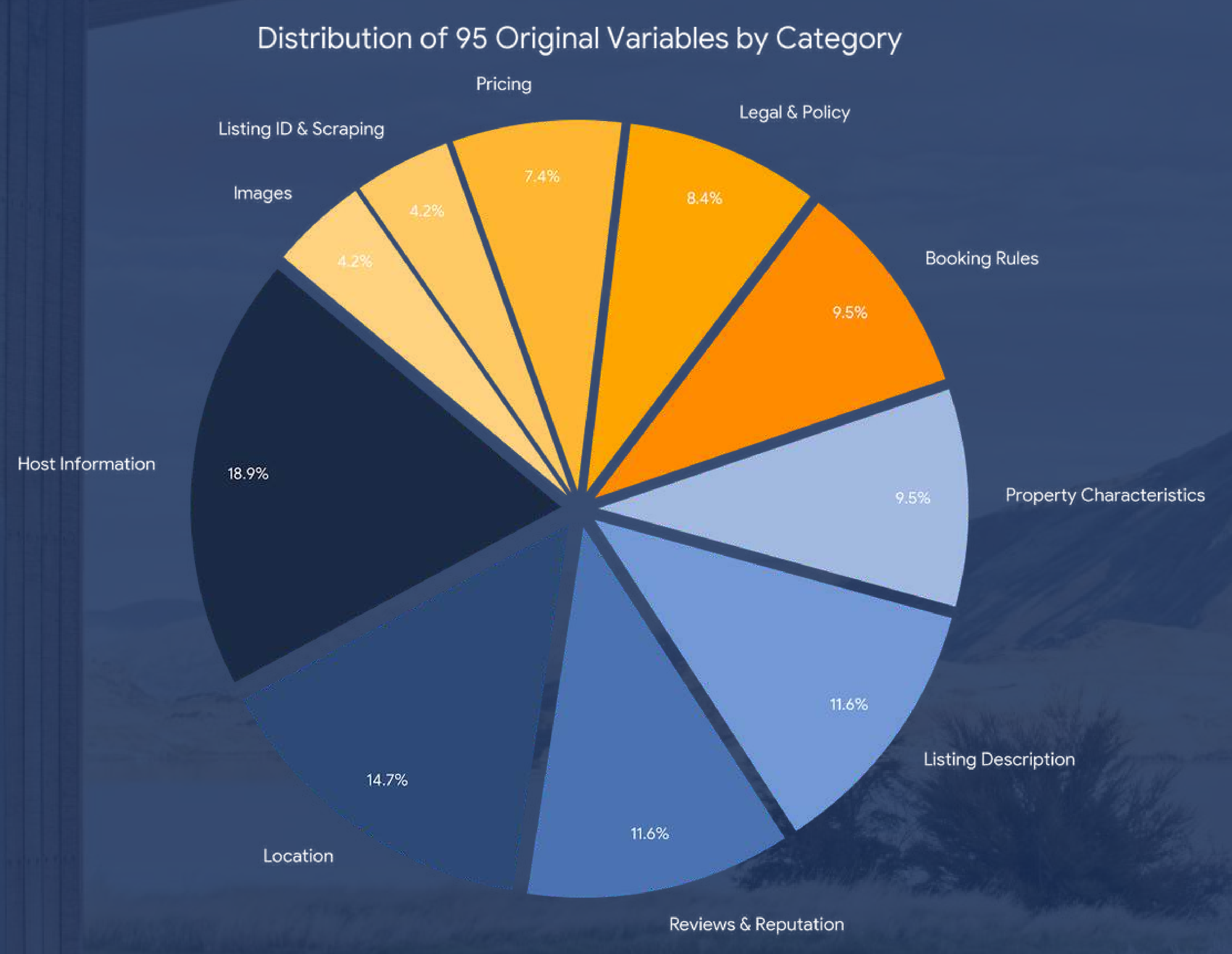
LISTING ATTRIBUTES: Room type, bed count, bed rooms count, property type, bathroom count, ACCOMMODATES, amenities count

LOCATION: Neighborhood, longitude/latitude

HOST PROFESSIONALISM: Response rate, Superhost status, acceptance rate, response time, host LISTING counts, identity verification.

BOOKING RULES: Minimum stay, cancellation policy, instant booking, extra people, guest included.

REPUTATION: Review scores



DATA CLEANING

DATA PREPARATION OVERVIEW

Raw Data → Inspection → Cleaning → Feature Engineering → Clean Dataset

Dataset

- 3,080 listings
- 95 variables

DATA CLEANING

Data Inspection

Descriptive text / metadata → removed

Variables with 98–100% missing values → removed

Review score variables and reviews_per_month contain missing values (~20–23%) → observations without ratings dropped

No duplicate observations detected

DATA CLEANING

variables converted

Monetary variables *(price, extra_people)* removing currency symbols and converted to numeric.

Percentage variables *(host_response_rate, host_acceptance_rate)* converted to numeric

Binary variables *(host_is_superhost, host_identity_verified, instant_bookable)* were converted to 0/1 indicators.

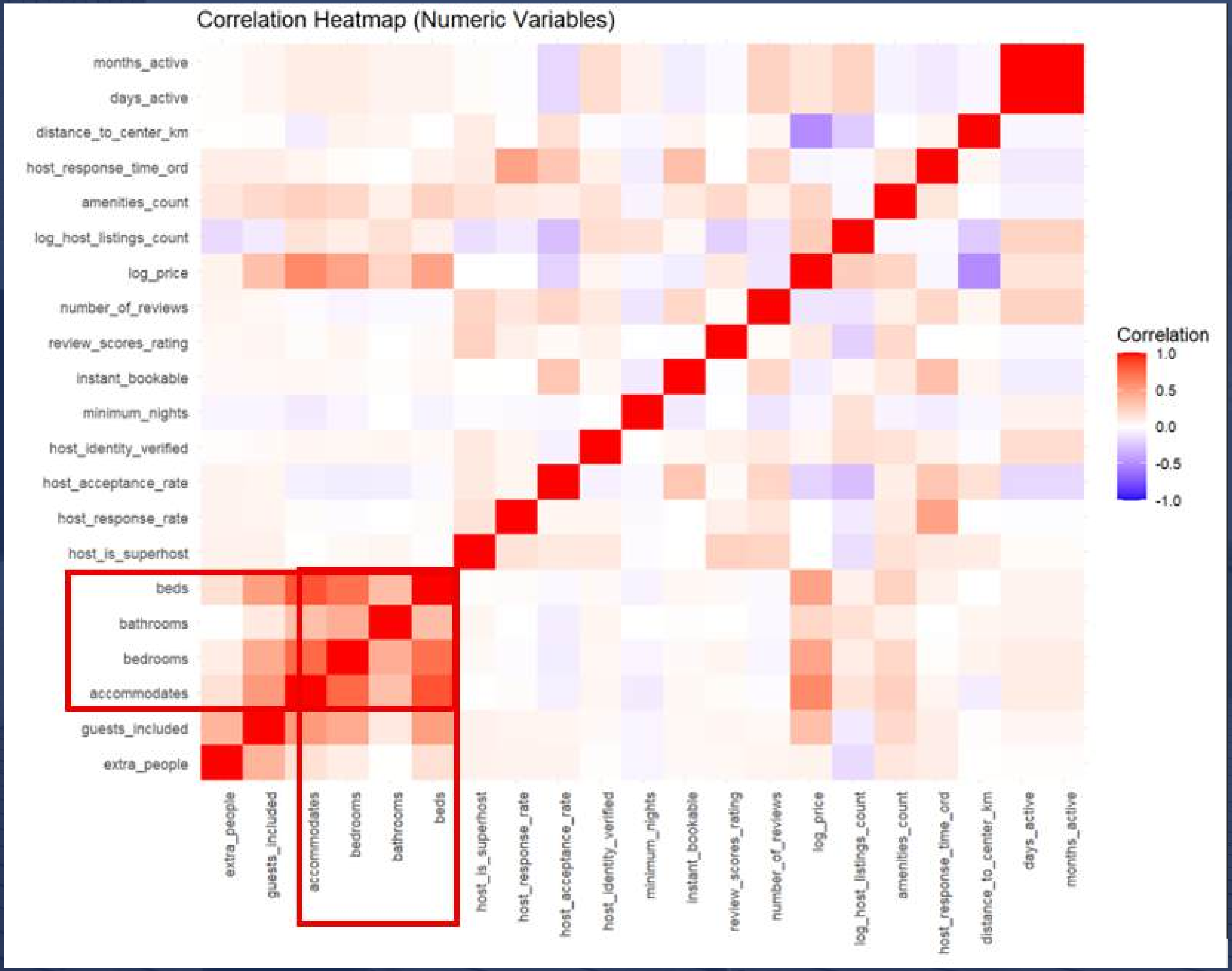
Date variables *(host_since, last_scraped)* were converted to Date format.

DATA CLEANING

FEATURE ENGINEERING

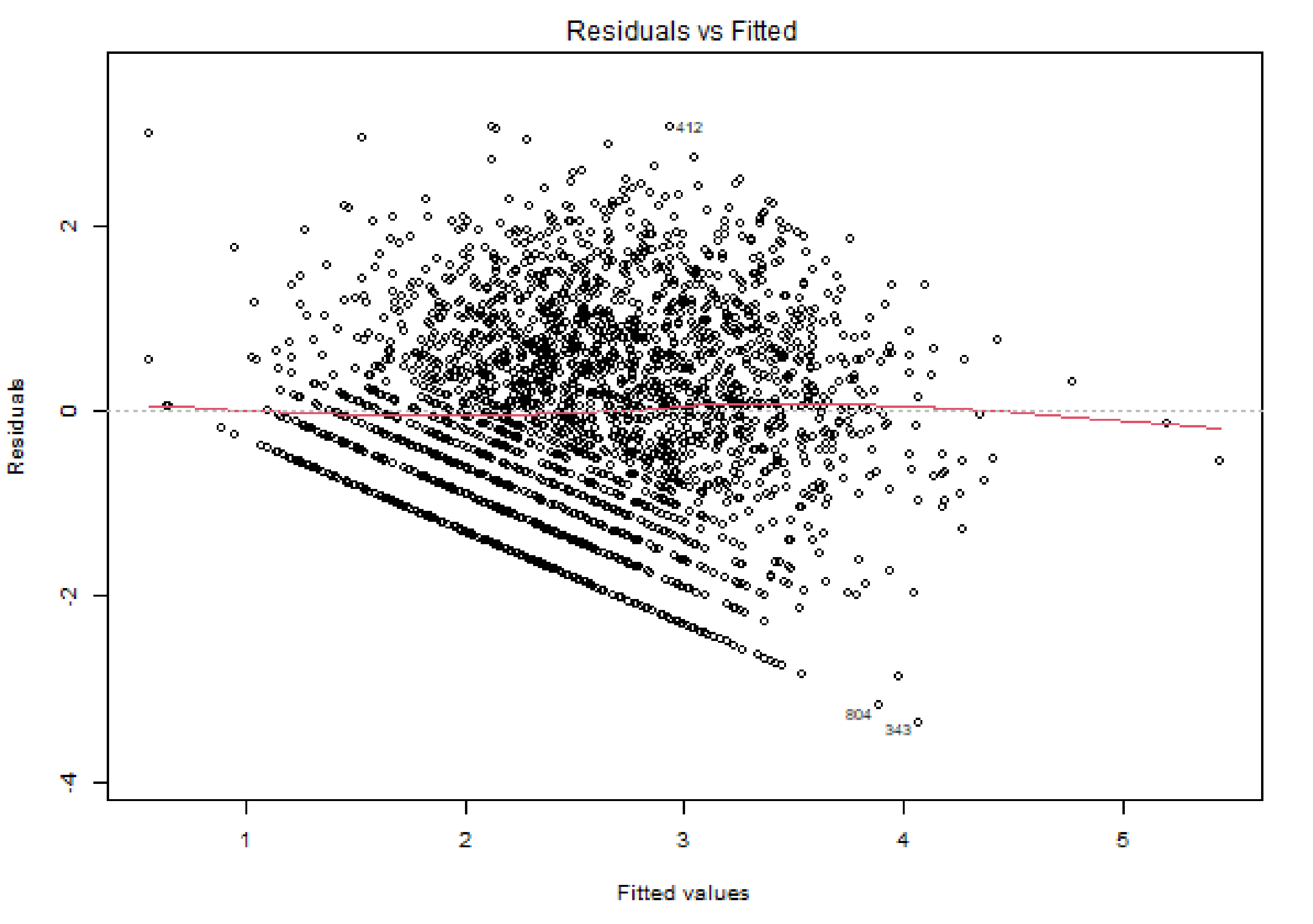
Raw Variable	Transformation	Econometric Objectives
Location (lat/long)	distance_to_center	Capture spatial accessibility
Amenities (text)	amenities_count	Quantify property quality
Reviews/month	months_active	Exposure variable, using host since - last scraped
Price	log(price)	Elasticity interpretation
Host listings	log(1+count)	Diminishing marginal effects
Minimum nights	Capped at 30	Reduce outlier influence
Host response time	Ordinal scale (1–4)	Ordered professionalism proxy

DESCRIPTIVE ANALYSIS



MULTIPLE LINEAR REGRESSION (MLR)

We tried using MLR, but...



Homoscedasticity is heavily violated

Also:

- Y variable has count characteristics
- YX relationship shouldn't be additive

KEY TAKEAWAYS:

MLR assumptions are violated
Results suggest transition to a count-based GLM models

Also, we are trying to see if our variables are justified, so:

Kitchen Sink	Simplified
Complete Predictors	Core Drivers Only

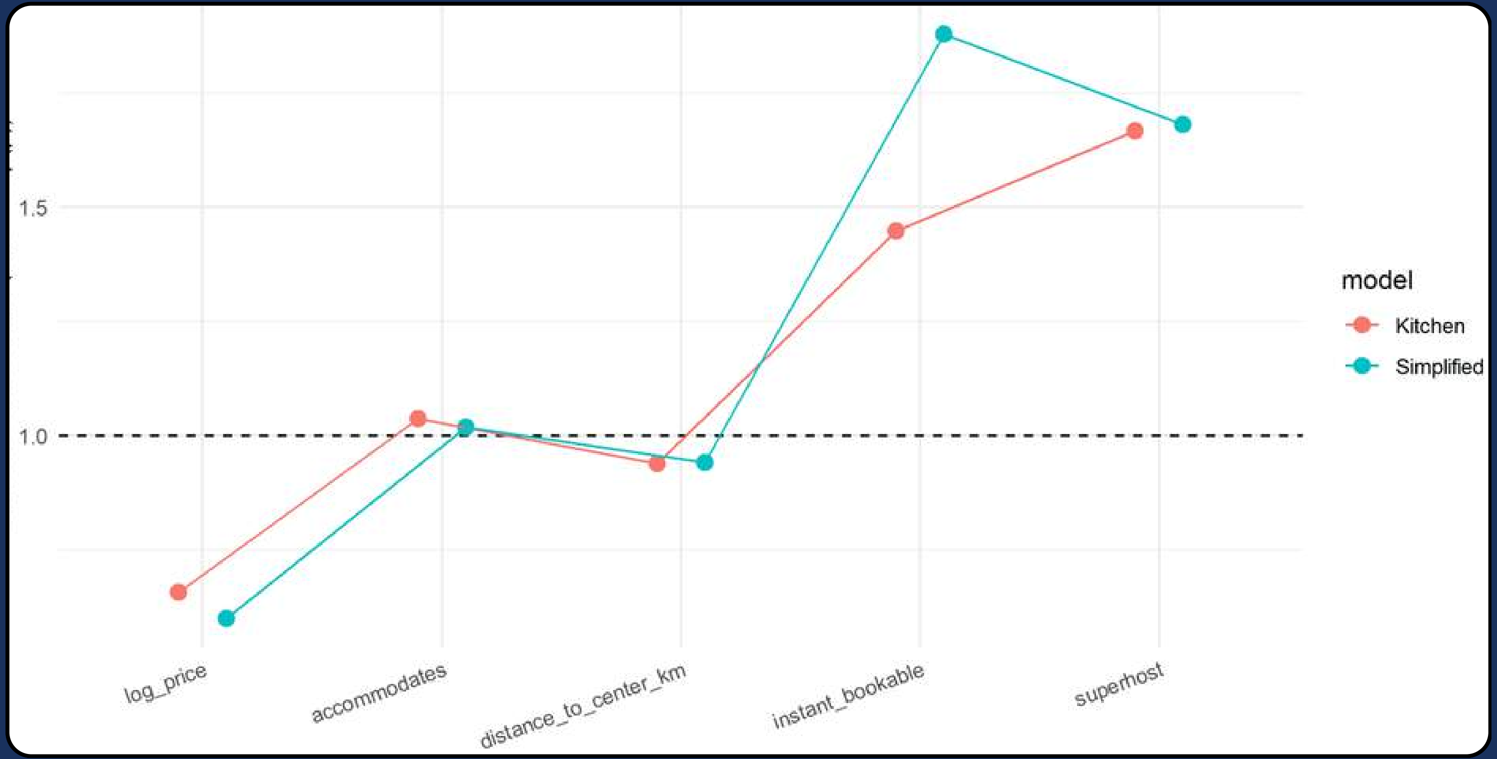
...for the next model

GLM | POISSON

MODELS COMPARED

Comparison	Kitchen Sink	Simplified
N	2513	
LRT (p-value)	<2.2e-1	
AIC	68,866.07	73,377.37

IRR Comparison Graph of Key Variables



The numbers favor the kitchen sink, but....

PEARSON DISPERSION

Kitchen Sink	30.4
Simplified	34.1

KEY TAKEAWAYS:

Poisson variance assumption is severely violated.

Results suggest transition to Negative Binomial model

GLM | NEGATIVE BINOMIAL

...PROBLEM SOLVED

Comparison	Poisson	Negative Binomial
Dispersion	~30	~1
AIC	68,866.07	20,375.7

*NB allows for overdispersion and provides a **better fit***

ANOTHER PROBLEM

Accomodates	+0.108
Beds	-0.079
Bedrooms	-0.051
Bathrooms	+0.039

beds and bedrooms produced counter-intuitive signs once controlling for accommodates

VARIABLE SOLUTION

Drop
- Beds
- Bedrooms
- Bathrooms

Keep
- Accomodates

Accomodates as primary size/capacity indicators

we think it's because:

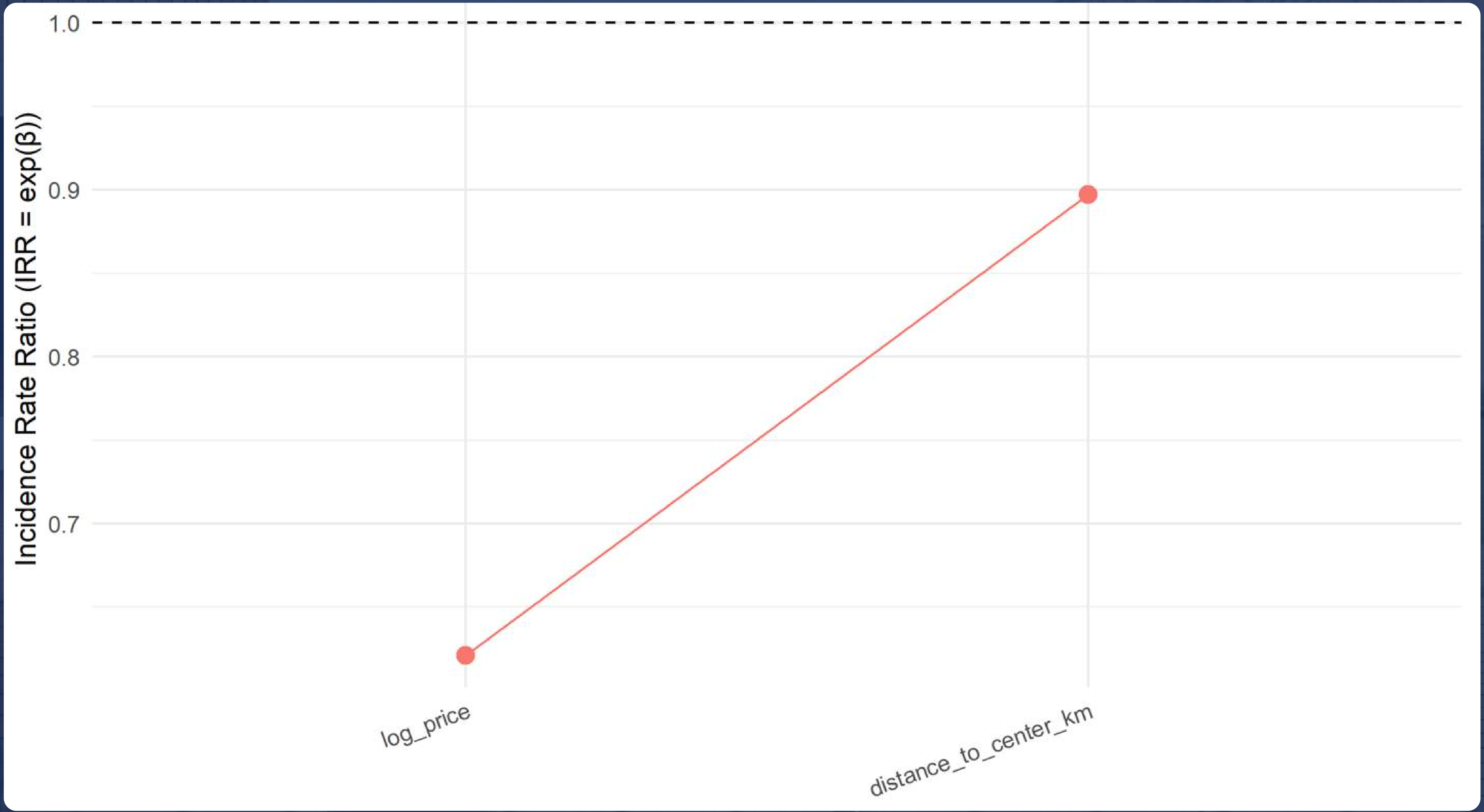
Beds
- Reduce privacy
- Reflect cheap configuration

Bedrooms
- Too much segmented layouts
- Reduce shared/common space

It does not means that more beds or bedrooms are worse

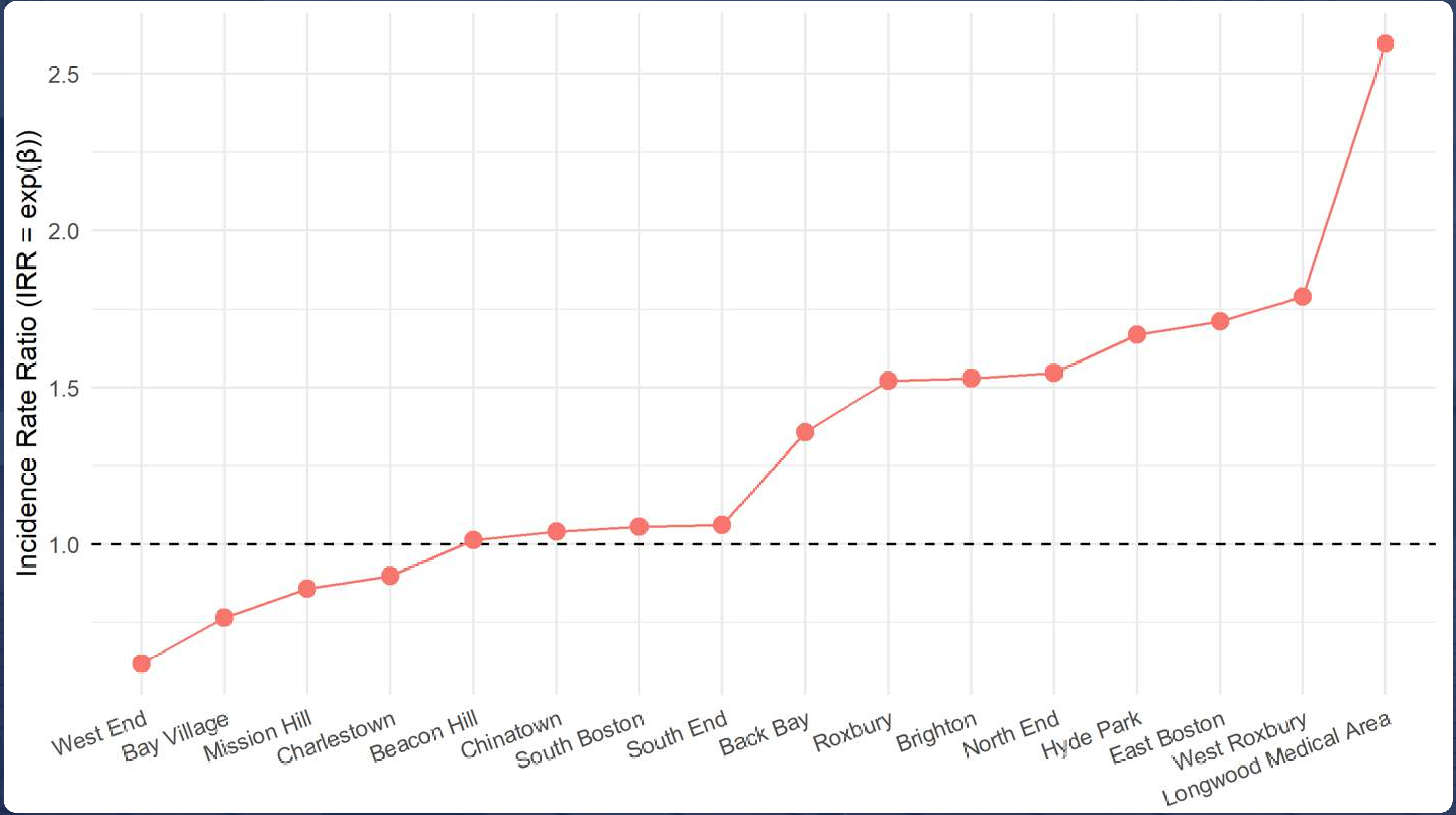
GLM | NEGATIVE BINOMIAL

RESULT
1. PRICE AND LOCATION(A)



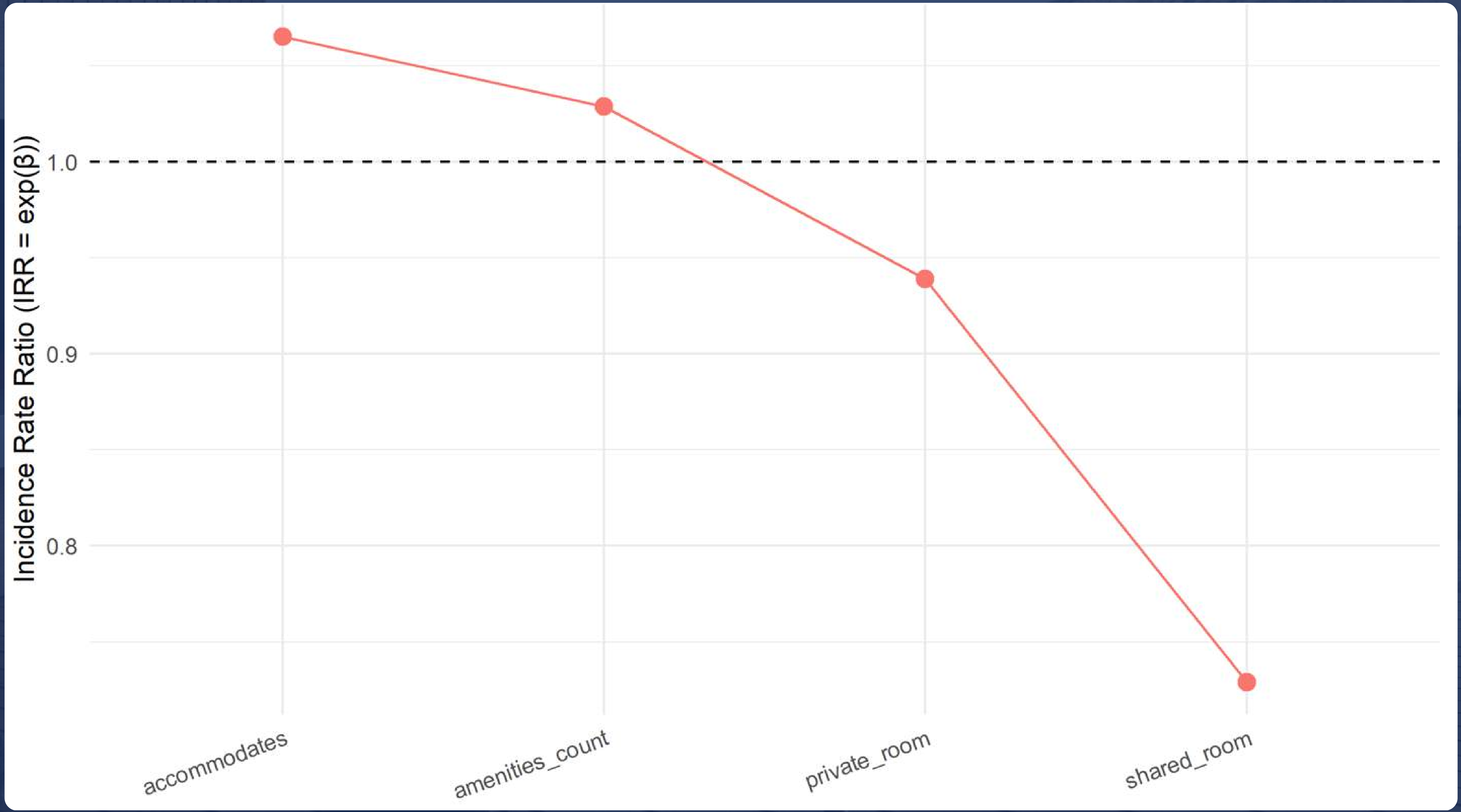
GLM | NEGATIVE BINOMIAL

RESULT
1. PRICE AND LOCATION(B) - NEIGHBORHOOD



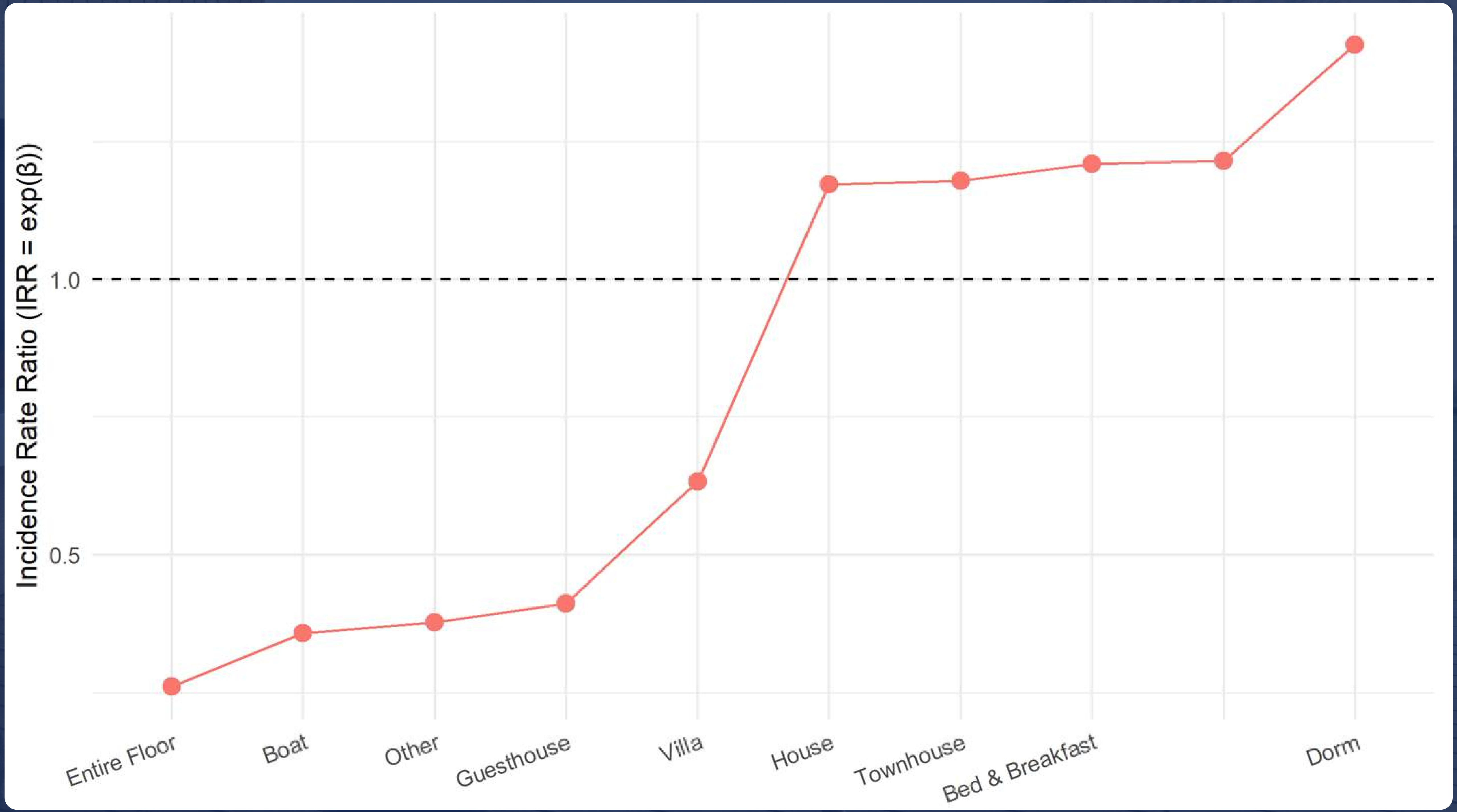
GLM | NEGATIVE BINOMIAL

RESULT
2. LISTINGS ATTRIBUTES(A) - CORE ATTRIBUTES



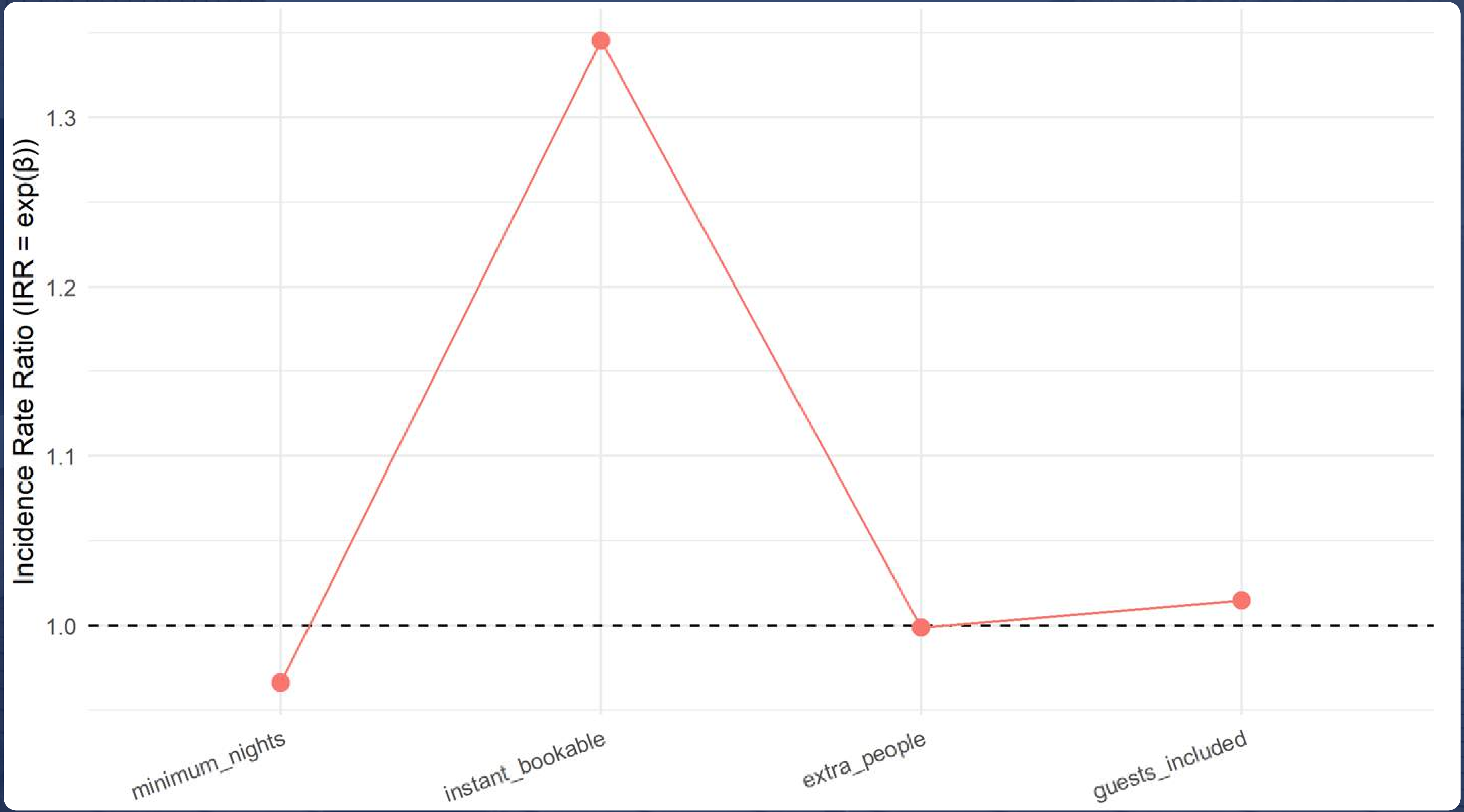
GLM | NEGATIVE BINOMIAL

RESULT
2. LISTINGS ATTRIBUTES(B) - PROPERTY TYPE



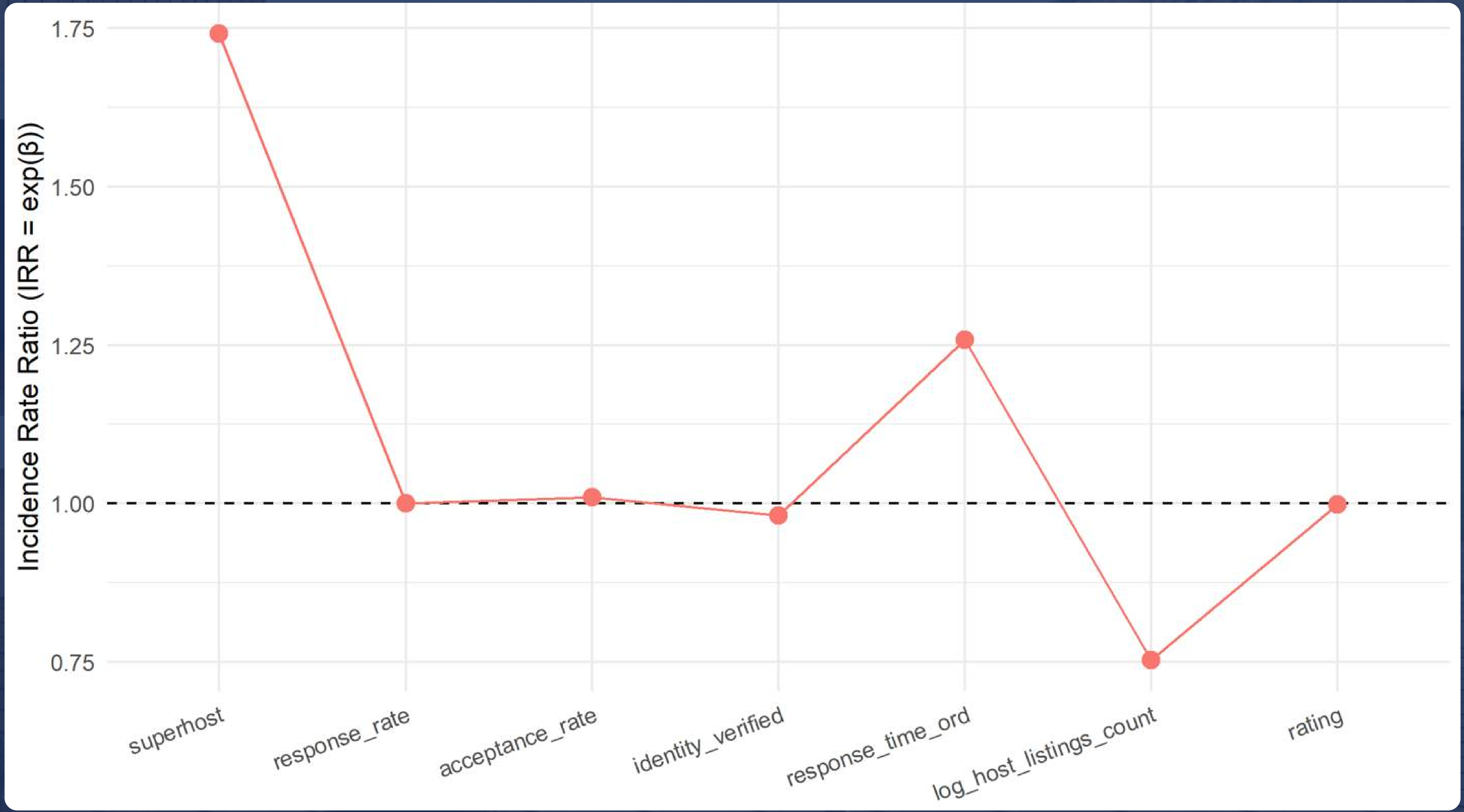
GLM | NEGATIVE BINOMIAL

RESULT
3. BOOKING RULE



GLM | NEGATIVE BINOMIAL

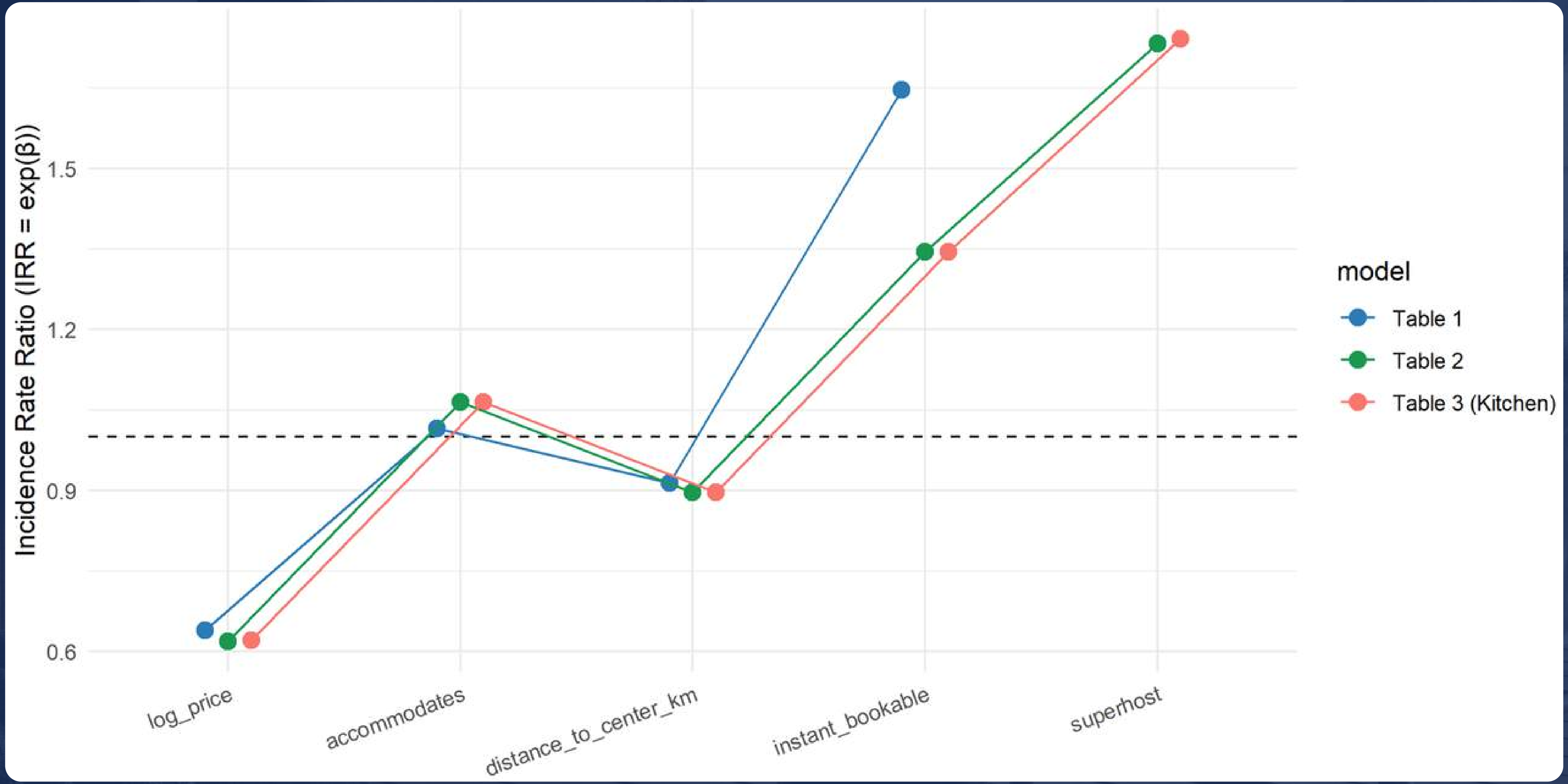
RESULT
4. HOST PROFESSIONALISM + REPUTATION



GLM | NEGATIVE BINOMIAL

LADDER STABILITY

IRR Comparison



Other Comparison

<table>
<tr><th>Variables</th><th>Table 1</th><th>Table 2</th><th>Table 3</th></tr>
<tr><td>θ (Theta)</td><td>0.897</td><td>1.077</td><td>1.077</td></tr>
<tr><td>AIC</td><td>20911</td><td>20374</td><td>20376</td></tr>
<tr><td rowspan="2">LRT (p-value)</td><td colspan="2">~0</td><td></td></tr>
<tr><td></td><td colspan="2">0.709</td></tr>
</table>

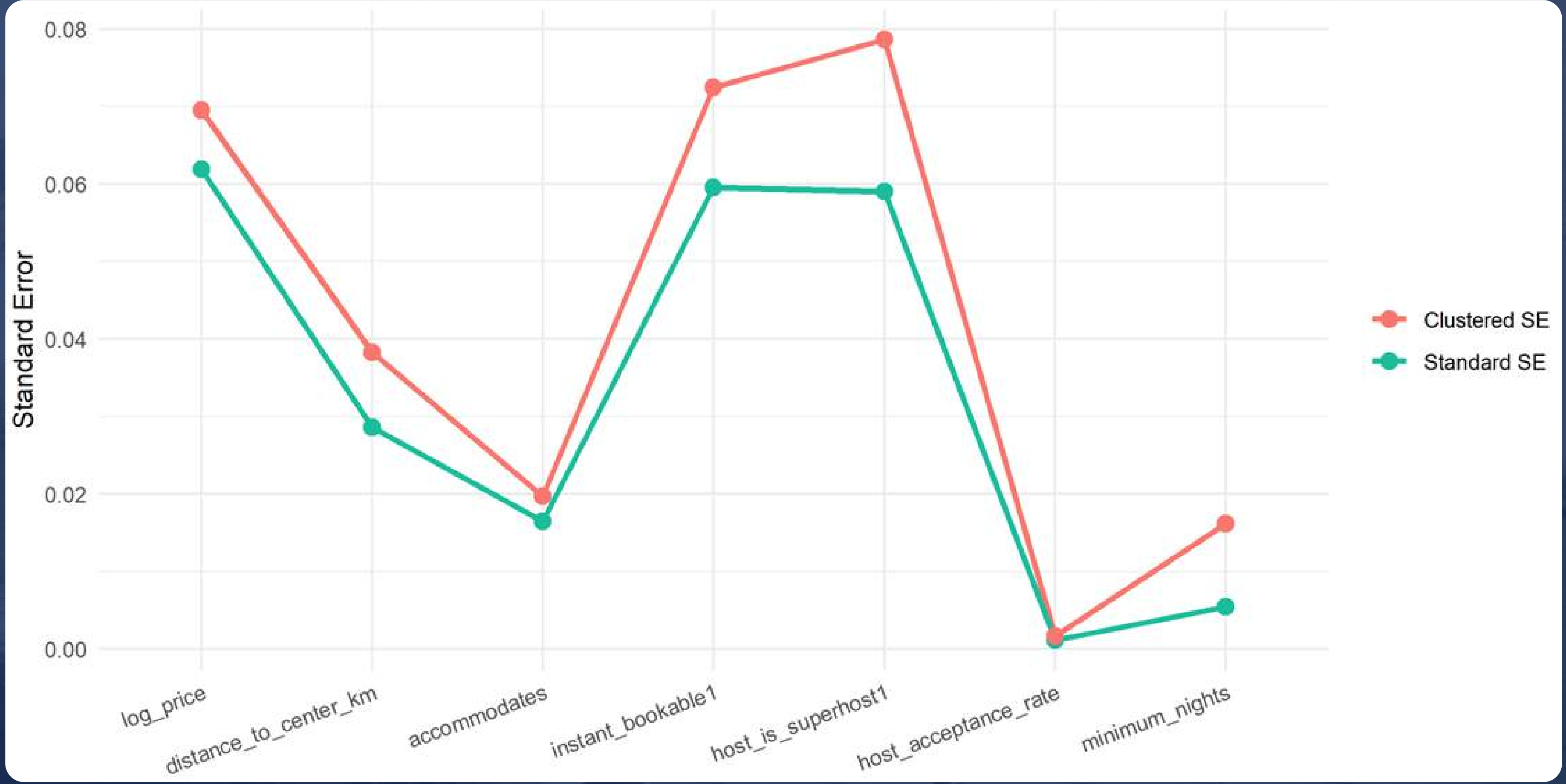
GLM | NEGATIVE BINOMIAL

ADDITIONALS: HOST CLUSTER

Multiple listings belong to the same host and may share unobserved characteristics (pricing strategy, management quality, branding).

INDEPENDENCE ASSUMPTIONS MAY BE VIOLATED

Variables	p-Value
log_price	<0.001
distance	0.004
instant_bookable	<0.001
superhost	<0.001
acceptance_rate	<0.001
minimum_nights	0.033



CONCLUSION

SE changed, but p-values are still acceptable (<0.05)

Main competitiveness drivers remain strong and statistically significant even after clustering.

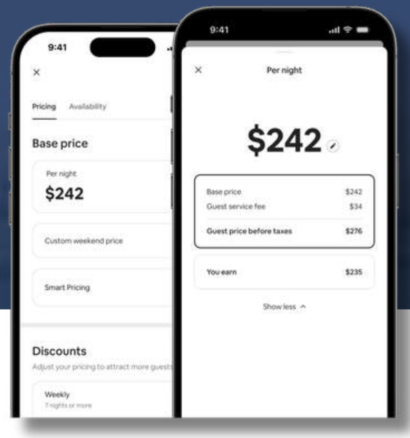
LIMITATION

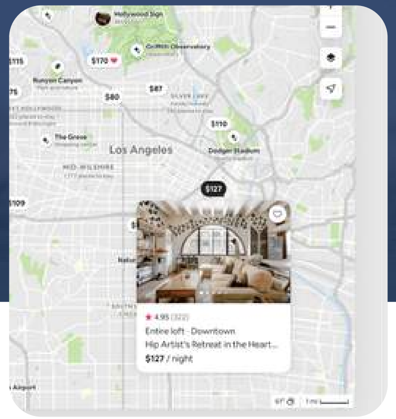
1. The analysis shows **relationships**, not causation
2. Potential **endogeneity** (pricing and host behavior)
3. **Host Cluster effect**
4. **Exposure proxy**
5. Model does not capture dynamic or **seasonal adjustments**
6. Reviews are used as a proxy for demand; **they do not directly measure bookings or revenue**

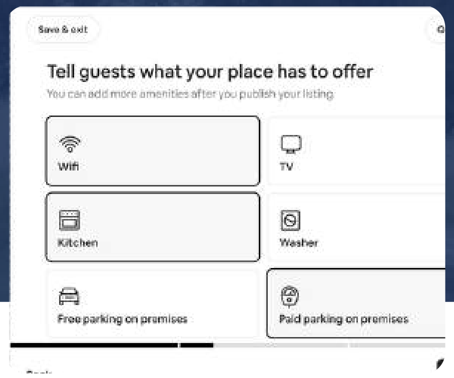
In future work, we could move closer to causal conclusions using better data and stronger methods

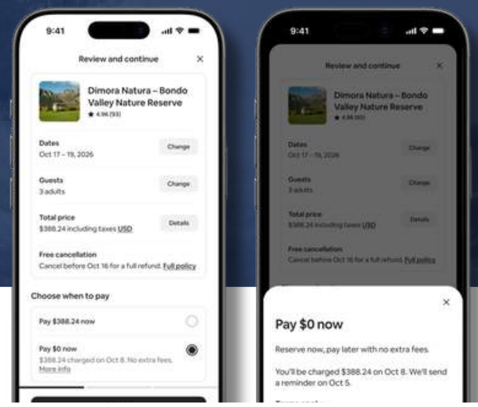
CONCLUSION

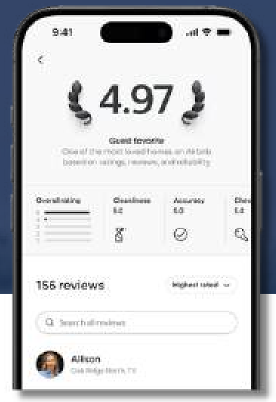
Airbnb is no longer a quality race, it’s **a competition beyond the baseline**











	PRICING	LOCATION	LISTINGS ATTRIBUTE	BOOKING RULE	HOST PROFESSIONALISM	REPUTATION
STRONG	**Price**	**Distance to Center**		**Instant Bookable**	**Superhost Status**	
MODERATE			**Amenities**	**Minimum Nights**	**Host Response Time, Host Acceptance Rate**	
WEAK				**Guests included, Extra people**	**Response rate, Identity verification**	**Rating**

BOSTON airbnb
LISTINGS ANALYSIS

THANKYOU

for your attention